

Innovate for Impact

AI was integrated throughout the course as a support tool for ideation, research, analysis, and design — with a distinctive use as a devil's advocate to challenge student thinking and encourage critical reflection.



COURSE AT A GLANCE

A project-based Bachelor elective bridging marketing strategy with real-world business challenges

5

ECTS CREDITS

70

STUDENTS

5th

SEMESTER

BA

BACHELOR LEVEL

COURSE OVERVIEW

Innovate for Impact

UNIVERSITY

**Amsterdam University of Applied
Sciences**

PROGRAMME

Marketing major

COURSE LEVEL

Bachelor · Semester 1 · 5 ECTS

TYPICAL STUDENT BACKGROUND

As this subject is designed to bridge academic theory with real-world business challenges, students must in teams (work in groups) to provide strategic (think out of the box), evidence-based marketing (research skills) and sales advice to companies, focus-ing on innovation and measurable value creation.

PREREQUISITES

Completion of all specialisation courses in the International Marketing & Sales Major.

SUMMARY OF AI USAGE



AI was integrated throughout the course as a support tool for ideation, research, analysis, and design. Students used tools such as ChatGPT, ResearchRabbit, Gemini, Perplexity, Canva, and Qualtrics to develop marketing concepts, conduct research, analyse data, and create prototypes. A distinctive element was the use of AI as a “devil’s advocate” to challenge ideas and encourage reflection. The main goal was to enhance creativity and efficiency while keeping students critical and responsible in their use of AI.

— ANTJE VAN ESSEN-HERMANS, COURSE TEACHER

SECTION 02

02

Learning outcomes

What the course set out to teach — and where AI made a difference.

INTENDED LEARNING OUTCOMES

Seven outcomes spanning creative thinking, research, design, and responsible AI use

01 Imagine creative marketing opportunities with appeal to a target audience.

03 Develop a SMART method to approach complex marketing problems within a given timeframe.

05 Design sustainable solutions to complex marketing problems by using the intervention cycle.

07 Use AI in the design and assessment of marketing solutions.

02 Define a complex marketing problem in an international business setting.

04 Integrate insights from academic and professional sources in the analysis of complex marketing problems.

06 Demonstrate an evidence-based line of reasoning using primary and secondary data.

AI SUPPORT · ILOs 1 & 2

AI as ideation engine and problem-definition partner

01

**INTENDED
LEARNING OUTCOME**

ILO 1: Imagine creative marketing opportunities with appeal to a target audience. ILO 2: Define a complex marketing problem in an international business setting.

“

AI supported creativity by helping students generate and test ideas for sustainable category extensions. It helped structure problem statements, research questions, and planning.

AI SUPPORT · ILOs 3 & 4

Structuring research and integrating academic sources with AI

03

**INTENDED
LEARNING OUTCOME**

ILO 3: Develop a SMART method to approach complex marketing problems within a given timeframe.
ILO 4: Integrate insights from academic and professional sources in the analysis of complex marketing problems.

“

AI tools also supported literature review, survey design, data analysis, and visual prototyping.

AI SUPPORT · ILOs 5 & 6

Prototyping sustainable solutions and building evidence-based arguments

04

**INTENDED
LEARNING OUTCOME**

ILO 5: Design sustainable solutions to complex marketing problems by using the intervention cycle. ILO 6: Demonstrate an evidence-based line of reasoning using primary and secondary data.

AI supported the design of sustainable marketing solutions through visual prototyping and data analysis. This included poster creation, intervention cycle templates, and survey data interpretation. It also helped students build evidence-based reasoning by assisting with data analysis and helping structure arguments grounded in primary and secondary research.

AI SUPPORT · ILO 7

Documenting AI use as a learning outcome in itself

07**INTENDED
LEARNING OUTCOME****Use AI in the design
and assessment of
marketing solutions.****“**

Throughout the course, students were asked to document and reflect on AI use, which supported critical evaluation and responsible application.

SKILLS ENHANCED

Breadth of skills built — and one area still needing work

“

AI especially supported creativity, research design, literature search, survey development, data interpretation, and visual communication. It also improved students' ability to structure complex tasks and work more independently. At the same time, the course showed that students needed support in evaluating output quality, identifying bias, and avoiding overreliance on AI.

AI-SPECIFIC LEARNING OUTCOMES

An explicit AI learning outcome — and a documentation requirement

YES

“

One explicit learning outcome is that students use AI in the design and assessment of marketing solutions. In addition, students are expected to document their AI use and reflect critically on how AI contributed to, limited, or biased their work.

SECTION 03

03

Purpose & benefits

Why bring AI into this course — and what students and lecturer got out of it.

MAIN MOTIVATION FOR USING AI

Creativity, efficiency, and critical professional practice

“

The main reason was to support creativity, research, and design in a real-world marketing project. AI was used to help students work more efficiently and to develop their ability to use AI critically in professional practice.

BENEFITS FOR STUDENTS

Faster, more accessible — and better at questioning AI outputs

“ AI helped students **generate ideas faster**, **structure research tasks**, and **create prototypes** more easily. It increased engagement and made some parts of the course more accessible, especially for students who needed support with writing, design, or research logic. It also encouraged experimentation and discussion. At the same time, students learned that AI works best when combined with human judgment and reflection.

BENEFITS FOR THE LECTURER

Sharper feedback loops — with extra investment in guidance

“

AI helped lecturers give clearer examples, support student ideation, and make discussions more concrete. It also improved efficiency because students could produce draft ideas, analyses, and visual outputs more quickly, which made feedback more focused. However, lecturers also needed extra time to guide students in prompting, evaluation, and responsible use.

SECTION 04

04

Implementation details

How AI showed up in class — the tools, the access model, and the guidance around them.

HOW AI WAS USED IN CLASS

Woven through every phase — from brainstorm to final presentation

“

AI was used across different phases of the course. In the early weeks, students used AI for brainstorming category extensions, team activities, and problem definition. Later, they used it for literature search, research planning, SWOT analysis, survey design, data analysis, poster creation, and final presentations. AI use took place in lectures, workshops, group work, and individual reflection tasks.

TOOLS & PLATFORMS

Eight tools across generative AI, research, and creative production

01

GENERATIVE AI

ChatGPT · Gemini · Claude · Perplexity

Used across all phases — brainstorming, problem definition, literature review, survey design, data analysis, and reflection tasks. Most were optional rather than compulsory.

02

RESEARCH & SURVEY

Research Rabbit · Qualtrics

Research Rabbit supported literature search and source mapping; Qualtrics was used for survey design and data collection as part of primary research phases.

03

CREATIVE & VISUAL

Canva · Kittl

Used for creating posters, prototypes, and visual outputs across design and communication phases of the project.

Most tools were used in free or freemium versions. Within each group, some students had paid subscriptions, which gave mixed access to advanced features.

ACCESS TO TOOLS - FREE VS PAID

Mixed access — free and paid versions used across student groups

Free

RECOMMENDED FOR ALL
STUDENTS

Free versions available for all core tools —
paid subscriptions used optionally within
student teams.

“

Students mainly used a combination of free and paid versions of tools. In each group there would be students with paid versions.

GUIDANCE & SUPPORT FOR USING AI EFFECTIVELY

Practical, ongoing — from demos to reflection tasks

“

Students received presentations on available platforms, demonstrations of prompting, in-class coaching, and feedback on their outputs. Reflection tasks were also used to help students think critically about bias, quality, and reliability. Support was practical and ongoing, especially when students were new to specific tools.

AI IN STUDENT ASSESSMENT

Rubric-based, documentation-required — AI use fully embedded

YES

Part of formal assessment rubrics

“

AI usage was part of the assessment through rubric-based criteria. Students were expected to use at least two AI tools, document their interactions, and reflect critically on how AI contributed to creativity, research, and design.

SECTION 05

05

Evaluation & risk management

How impact was measured — and how misuse was kept in check.

HOW AI IMPACT WAS EVALUATED

AI usage logs, instructor reflection, coaching feedback, and planned surveys

“

The impact was evaluated through student AI usage logs, instructor reflections, feedback during coaching sessions, and review of student outputs. The course also planned pre- and post-course AI literacy surveys by AI HED team.

METRICS & KPIS

Defined indicators — engagement, reflection quality, and AI helpfulness

YES

“

Suggested indicators included student performance, engagement, quality of AI reflections, total amount of students reporting AI as helpful, number of AI-generated ideas that were used or revised.

METRICS & KPIS

Defined indicators — engagement, reflection quality, and AI helpfulness

Statement	Mean	N
This course helped me to use AI in a responsible way, following ethical considerations and privacy rules.	3.72	39
This course helped me to evaluate AI-generated answers critically.	4.00	38
This course helped me to use AI effectively in the course topic(s).	4.10	39
The integration of AI in this course improved my learning experience.	3.84	38
The use of AI in this course helped me better understand complex concepts.	4.11	38
The use of AI in this course helped me apply knowledge in practical ways.	4.08	38
The teacher guided us in using AI in a responsible way, following ethical considerations and privacy rules.	3.41	39
The teacher guided us in evaluating AI-generated answers.	3.71	38
The teacher guided us in using AI effectively.	3.59	39
The integration with AI in this course helped me develop relevant skills for my future.	4.03	37
I feel more prepared to use AI in other academic or professional contexts because of this course.	3.74	39
Overall course evaluation	3.85	

FORMAL COURSE EVALUATION

Embedded in assignments, logs, and assessment rubrics

YES

Part of the course's formal assessment activities

“

AI use was embedded in the course through assignments, reflection logs, and assessment rubrics. It was not treated as a separate activity, but as part of the regular evaluation of student work and course development.

RISK MANAGEMENT IN AI USE

Five risks identified — managed through guidance and reflection

YES

Structured from the start of the course

“ The main risks identified were overreliance on AI, uneven AI literacy, bias in outputs, superficial reflection, and technical or access issues. These were managed through lecturer guidance, peer support, structured prompting exercises, reflection tasks, and clear expectations that AI should support rather than replace student thinking. Students were encouraged to validate outputs and compare them with theory, evidence, and their own judgment.

SECTION 06

06

Overall impact

What changed — for the teaching, the learning, and the student work.

OVERALL IMPACT ON TEACHING & LEARNING



Overall, AI had a positive impact on both teaching and learning. It **increased engagement, supported creativity,** and made several tasks more **efficient**, especially in ideation, research, and design. It also helped students participate more actively and made some complex tasks feel more accessible.

At the same time, the course showed that AI only adds value when students are guided to **question outputs, reflect critically, and use the tools selectively**. Challenges for lecturers included managing different student skill levels, supporting tool use, and dealing with technical limitations.

— ANTJE VAN ESSEN-HERMANS, COURSE TEACHER

IMPACT ON STUDENT LEARNING OUTCOMES

Strong gains in creativity and efficiency — critical reflection still needed work

“

AI supported student learning outcomes in a mainly positive way. It improved idea generation, research structuring, survey development, and visual communication. Students also became more aware of bias, prompting, and the need to evaluate AI outputs. The strongest gains were in creativity, efficiency, and engagement, while the weakest area remained critical reflection, which still needed continuous support.

MAIN CHALLENGES & HOW THEY WERE MANAGED

Uneven experience, overreliance, and tool overload

“

The main challenges were uneven student experience with AI, overreliance on outputs, limited free access to some tools, technical issues, and difficulty judging output quality. Some students also felt overloaded by the number of available tools. These challenges were managed through demonstrations, in-class coaching, peer learning, reflection logs, and by encouraging students to use AI as a co-creator rather than as a ready-made answer machine

SECTION 07

07

Scalability & transferability

How this experience travels to other courses — and what it would take to support that.

SCALING TO OTHER COURSES AND NEED FOR INSTITUTIONAL SUPPORT

Yes — especially for project-based courses

100%

YES — SCALABLE

“

This experience can be scaled to other courses, especially project-based courses where students need support in ideation, research, analysis, and communication. To scale it well, lecturers would need basic training in AI tools and prompting, clear assessment criteria, and practical guidelines for responsible use.

KEY ADVICE FOR EDUCATORS

Four principles for purposeful, student-centred AI integration

Introduce AI as support for thinking.**01**

Students benefit most when AI is introduced as support for thinking, not as a substitute for it.

Guide early — prompting, bias, and reflection.**02**

Give students practical guidance early, especially on prompting, bias, and reflection.

Show good examples in class.**03**

Start with a small number of tools and connect them clearly to the learning outcomes.

Intervene when AI becomes a shortcut.**04**

Keep AI use purposeful and do not let the tool drive the course.

A course that made AI a co-creator — without letting it become a substitute for thinking.

COURSE

Innovate for Impact

INFORMANT

ANTJE VAN
ESSEN-
HERMANS

INSTITUTION

AUAS · AMSTERDAM

PROGRAMME

Marketing major

ABOUT AI-HED

An Erasmus+ cooperation project on AI in higher education

The project "**Artificial Intelligence in Higher Education Teaching and Learning**" (AI-HED), launched with co-funding from the European Commission under the Erasmus+ Programme in the fields of education, training, youth, and sport for the 2021–2027 period, was submitted and awarded under **Key Action 2: Cooperation among organisations and institutions**.

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